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ISOLATION, ENRICHMENT, AND EXPANSION OF HUMAN ROD PROGENITOR CELLS AND EXTRACELLULAR VESICLES DERIVED THEREFROM

Abstract

Described herein are methods for producing enriched populations of progenitor rod photoreceptor cells (PRPs) and EVs derived therefrom, as well as methods of using the PRPs and EVs to diagnose and treat retinal disease.

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Background/Summary

CLAIM OF PRIORITY [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/339,252, filed on May 6, 2022. The entire contents of the foregoing are hereby incorporated by reference.

TECHNICAL FIELD

[0002] Described herein are methods for producing enriched populations of progenitor rod photoreceptor cells (PRPs) and extracellular vesicles (EVs) derived therefrom, as well as methods of using the PRPs and EVs to diagnose and treat retinal disease.

BACKGROUND

[0003] Retinal degenerative diseases, which lead to the death of rod and cone photoreceptor cells, are the leading cause of inherited vision loss worldwide. Retinitis Pigmentosa, for example, is a disease in which rods photoreceptor die first leading to the death of cone photoreceptors. Induced pluripotent or embryonic stem cells (iPSCs/ESCs) have been proposed as a possible source of new photoreceptors to restore vision in these conditions. Studies carried out in mouse models of retinal degeneration over the past decade have highlighted several limitations for cell replacement or neuroprotection in the retina, such as poor integration of grafted cells in the host retina and high dosage needed to neuro-protect the retina.

SUMMARY

[0004] Described herein are methods of providing a population of progenitor rod photoreceptor cells (PRPs). The methods can include providing an initial population comprising retinal cells from a mammal, preferably a fetal mammal (alternatively, the initial population can comprise cells from a retinal or optic cup organoid generated from iPSC or ESC). Cells expressing CD73 and CD276 (and optionally Cd11b) are isolated from the initial population to provide a substantially purified population of PRPs.

[0005] In some embodiments, isolating cells expressing CD73 and CD276 (and optionally Cd11b) comprises using fluorescence activated cell sorting (FACS) or Magnetic-activated cell sorting (MACS), optionally in a microfluidic device. For example, the methods can include contacting the initial population of cells with antibodies that bind to CD73 and CD276, and isolating cells to which antibodies to both are bound.

[0006] In some embodiments, the methods further comprise culturing the initial population comprising retinal cells before isolating cells expressing CD73 and CD276 (and optionally Cd11b).

[0007] The methods can include optionally culturing the substantially purified population of CD73+/CD276+ PRPs obtained. In some embodiments, the substantially purified population of PRPs is cultured in media comprising serum or serum replacement, EGF, and FGF. In some embodiments, the substantially purified population of PRPs is cultured in hypoxic conditions (e.g., 3-10% O₂). In some embodiments the hPRPs are also CD11b+.

[0008] In some embodiments, the initial population comprising retinal cells is from a fetal mammal, e.g., a human.

[0009] Also provided herein are substantially purified populations of PRPs produced by a method described herein. In some embodiments, at least 75%, e.g., at least 80%, 85%, 90%, or 95%, of the population comprises cells expressing both CD73 and CD276 (and optionally Cd11b). Further

provided are the substantially purified populations of PRPs for use in treating a subject who has a condition associated with loss of retinal rod or cone photoreceptors, as described herein. In some embodiments, the condition associated with loss of retinal rod or cone photoreceptors is an inherited retinal degenerative disease (IRD), optionally cone-rod dystrophy, retinitis pigmentosa (e.g., LCA), or Stargardt's disease; or macular degeneration, e.g., dry macular degeneration. Also provided herein are compositions comprising PRPs as described herein (e.g., that are at least 75%, e.g., at least 80%, 85% 90%, or 95%, of the population comprises cells expressing both CD73 and CD276 (and optionally Cd11b)), optionally formulated in a physiologically acceptable buffer or polymeric gel, e.g., as described herein, optionally gelatin-hyaluronic acid (HA) hydrogel, e.g., a composition comprising gelatin hydroxyphenylpropionic acid (gelatin-HPA) and hyaluronic acid-tyramine (HA-Tyr).

[0010] Additionally, provided herein are methods for treating a subject who has a condition associated with loss of retinal rod photoreceptors. The methods can comprise administering to the subject (e.g., by injection) a therapeutically effective amount of PRPs obtained by a method described herein, preferably wherein at least 75%, e.g., at least 80%, 85%, 90%, or 95%, of the population comprises cells expressing both CD73 and CD276 (and optionally Cd11b), optionally formulated in a physiologically acceptable buffer or polymeric gel, e.g., as described herein, optionally gelatin-hyaluronic acid (HA) hydrogel, e.g., a composition comprising gelatin hydroxyphenylpropionic acid (gelatin-HPA) and hyaluronic acid-tyramine (HA-Tyr)).

[0011] Further, provided herein are methods of obtaining or providing extracellular vesicles (EVs) from rod photoreceptor progenitor cells. The methods can comprise providing a substantially purified population of PRPs as described herein; maintaining the substantially purified population of PRPs in culture in media, preferably wherein the media is in contact with the PRPs in the culture for at least 12, 18, 24, 36, 48, or 72 hours; removing the media from the culture; and isolating EVs from the media.

[0012] In some embodiments, isolating EVs comprises ultracentrifuging the media to obtain a pellet comprising EVs. An exemplary ultracentrifuge protocol comprises 3 steps: 1500 g for 30 min; 10,000 g for about 20 min; and 15,000 g for 30 min. In some embodiments, the methods further comprise lyophilizing the isolated EVs.

[0013] Also provided herein are compositions comprising EVs obtained by a method described herein. In some embodiments, the EVs are in a polymeric gel scaffold, e.g., a polymeric scaffold comprising polycaprolactone (PCL), polylactic-glycolic acid (PLGA), polyethylene glycol (PEG), polylactic acid (PLA), polyetherimide (PEI), PNIPAAm, or hydroxy ethyl-methacrylate. Also provided are the compositions for use in treating a subject who has a condition associated with loss of retinal rod or cone photoreceptors. Additionally provided herein are methods for treating a subject who has a condition associated with loss of retinal rod or cone photoreceptors, as described herein. The methods can comprise administering to the subject a therapeutically effective amount of PRPs obtained by a method described herein.

[0014] In some embodiments, the condition associated with loss of retinal rod or cone photoreceptors is an inherited retinal degenerative disease (IRD), optionally cone-rod dystrophy, retinitis pigmentosa (e.g., LCA), or Stargardt's disease; or macular degeneration, e.g., dry macular degeneration.

[0015] As used herein, unless otherwise specified the term "about" mean $\pm 10\%$ of the indicated number.

[0016] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Methods and materials are described herein for use in the present invention; other, suitable methods and materials known in the art can also be used. The materials, methods, and examples are illustrative only and not intended to be limiting. All publications, patent applications, patents, sequences, database entries, and other references mentioned herein are incorporated by reference in

their entirety. In case of conflict, the present specification, including definitions, will control.
[0017] Other features and advantages of the invention will be apparent from the following detailed description and figures, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0018] FIG. 1. Summary of exemplary isolation methods for human progenitor rod photoreceptors (hPRP). Cells are isolated from human fetal retina with CD73+/CD276+ markers. Post-sorting cells show high purity >95% for both markers along with other specific rod expressing markers. High viability can be seen both post-sorting and after weeks of cell culture with media as described herein. The markers shown demonstrate maturation and cell lineage.

[0019] FIG. 2. Diagram of exemplary neuroprotective human EVs extracted from hPRP. Illustrative exemplary diagram showing multiple encapsulated molecules and proteins in EVs along with transmembrane proteins enabling EVs signaling and genesis.

[0020] FIG. 3. Illustrative schematic of exemplary EVs manufacturing process. Manufacturing process of extraction of EVs from hPRP along with encapsulation and in vivo injection for potential neuroprotective effect on retinal diseases.

[0021] FIG. 4. Total number of cells throughout the process of sorting. Total number of cells in function of time (weeks of culture). Number of cells was measured in T75 flasks using Trypan blue and hemacytometer. The average number of cells after digestion of retina is 2 million and is cultured until reaching 10 million cells. Upon reaching this number cells are sorted; therefore, total number drops down. Then the isolated rod progenitor cells are cultured until reaching high number (40 million) of cells at 5 weeks.

[0022] FIG. 5. Brightfield images of rods hPRP post-sorting in T75 flasks. Cells were imaged with Leica Brightfield upright microscope showing successful cell attachment, viability, morphology, and proliferation over 20 days of culture.

[0023] FIG. 6. Live sorting and gating with a MACSQUANT cell sorter using a TYTO cartridge (Miltenyi), Clive CD73 and CD276 gating of fetal retinal cells to isolate pure hPRP. A clear population was seen for both CD73 and CD276 and can be isolated with high viability and purity post-sorting.

[0024] FIG. 7. TYTO Sorting data analyzed for input, output and negative fractions of cells. Expression of sorting markers (CD73 and CD276) during the sorting process for the input, output and negative fraction of cells. Data was analyzed using the TYTO sorter and the MACSQUANT flow cytometer. It shows the expression of CD73/CD276 positive cells before and after the first sort. CD73/CD276 population starts around 30-40% in the unsorted sample and rises to 95% after sort. Data is shown as mean±SD for 10 replicates (10 different sorts). Statistical one-way ANOVA was performed and shows significant difference between input and output (*p<0.001).

[0025] FIG. 8. Live timing and analysis of sorted cells with TYTO. Cell sorter analysis showing triggered event, sort rate and percent of positively sorted cells across a 1 h sorting. A constant >90% sorted can be observed for the entire sorting process.

[0026] FIG. 9. Raw sorting markers expression analyzed with live flow cytometry. Expression of sorting markers (CD73, CD276) with their specific fluorophore (APC, PE, VioBlue) before and after sort. Data was analyzed using the MACSQUANT flow cytometer. It shows data of the sort—CD73/CD276 used in two fluorophores. Top right panel line: scatter and histogram plot of CD73/CD276 expression for input. Middle right panel line: scatter and histogram plot of CD73/CD276 expression for output. Top left panel: gating strategy and population number corresponding to the gates. Histogram in both figures shows that, post sorting, CD73 and CD276 positive cell population was increase. This is proven by the higher intensity of the peaks of these

markers. A minimum of 50,000 cells was used for each sample in flow cytometry.

[0027] FIG. **10**. Flow cytometry data evolution for 16-weeks tissue during the process of sorting. Left side shows the gating strategy for cells, singles and DAPI positive events to analyze hPRP. Right side shows the expression of purity markers (Recoverin, CD73, CD276) for both unsorted and hPRP. Expression of those 3 rod markers was found to be extremely high and pure in hPRP while being between 20-40% present in the unsorted population. This suggests that our sorting strategy enables to capture a pure population of rod progenitor cells. A minimum of 50,000 cells was used for each sample in flow cytometry. Data is shown as mean $\pm$ SD for 5 replicates (5 different flow experiments).

[0028] FIG. **1**. Flow cytometry data for hPRP vs fetal retina. It shows the differences in all these markers between a sorted hPRP and unsorted fetal retina tissue (also known as hRPC). Overall, rod markers (CD73, CD276, Recoverin, Cone rod homeobox (CRX)) were found to be extremely high in hPRP compared to unsorted retina. Other markers of retinal cells (cones, RGC, bipolar, horizontal and amacrine) were found negative in hPRP. Finally, 15-20% of hPRP were found to be proliferative, with a 50% PAX6 retinal marker expression. A minimum of 50,000 cells was used for each sample in flow cytometry.

[0029] FIG. **12**. Immuno-staining analysis performed on hPRP with sorting markers. CD73, CD276 and CRX are distinct rod photoreceptors markers and hPRP were stained with same primary antibodies. High expression of these 3 markers can be observed in the IHC images. The nuclear stain used to identify the cells was DAPI.

[0030] FIG. **13**. Immune staining for different markers for hPRP. Cells were stained with common rod markers (Rhodopsin, Recoverin) along with retinal marker (PAX6) and proliferation marker (K167). Overall, all markers were found in hPRP. Other retinal markers like Calbindin (Amacrine cells), Recoverin (photoreceptors), PKCa (Bipolar) and Rhodopsin (rod photoreceptors) were tested, and expression varied from low to no expression. Nuclei were stained with DAPI and all images are taken at 20 $\times$ magnification.

[0031] FIG. **14**. Whole mounted retina of RD1 mouse post-transplantation. In normal cases, RD1 mice at P31 should not possess any cone or any ONL staining due to the fast degeneration process of the modified specie; this is what was observed in control samples. However, in animals which received an injection of hPRP, staining showed here was found. Cone Arresting positive staining was found which suggests that the injection of hPRP in the vitreous has a strong neuroprotective impact on the retina of RD1 mice.

[0032] FIG. **15**. ONL presence in RD1 mice post-transplantation. In normal cases, RD1 mice at P31 have extremely degenerating retina that can be seen as thinning of ONL or absences of one. To measure the impact of hPRP injection in this fast degenerative mouse model, we analyzed in a binary fashion the presence of one line of ONL nuclei and gave scores to all animals with multiple sections. As seen in the graph on the left, the score grade was found to be higher for animals which received hPRP injection in the subretinal space and even higher for the vitreous injection. This suggests that the injection of hPRP in the vitreous has a strong neuroprotective impact on the retina of RD1 mice.

[0033] FIG. **16**. Intraocular pressure (IOP) testing of RD1 mice. In normal cases, RD1 mice at P31 should possess an IOP between 10-20 mmHg, which was observed in the control eye. IOP was found to be increased by the injection of cells in the vitreous while no change was observed for control and subretinal injection. However, IOP in eye with vitreous injection was found to go back to the baseline as soon as 1-week post-transplantation. This study suggests a high safety for the injection of hPRP in the eye of rodents.

[0034] FIG. **17**. Comparison of Extracellular Vesicle Proteomic Cargo between our EVs and Developing Retina.

[0035] FIG. **18**. Summary of Vesicle Proteomic Cargo Composition.

[0036] FIG. **19**. Fold change of total proteome cargo in our EVs, segregated into the 6 categories

and subcategories.

[0037] FIG. 20. Fold change in protein expression comparison between our EVs and the fetal retina (using only a 5× limit).

[0038] FIG. 21. Proteins exclusive to our EVs and proteins absent from the fetal retina.

[0039] FIGS. 22A-B. Functional behavior testing in Rho^{-/-} mice by optomotor testing. Mice were injected with either (cell+gel) or SHAM (pure saline water) at 3 weeks and tested weekly for functional retinal behavior with OMR apparatus. A. Visual acuity (VA) of mice measured as threshold of spatial frequency. B. Contrast sensitivity threshold of mice, measured at 0.2 cycle/degree VA. (**p=0005).

[0040] FIGS. 23A-C. Cone retinal function testing in Rho^{-/-} mice by ERG. Mice were injected with either (cell+gel) or SHAM (pure saline water) at 3 weeks and tested weekly for cone retinal function with ERG apparatus. A. Photopic B-wave measured at 600 cd.s/m² for both groups. B. Percent increase calculation for Cell+Gel vs SHAM injections. C. Representative photopic B-wave traces at 6 and 7 weeks. (**p=0.005).

[0041] FIGS. 24A-B. Cone retinal function testing in P23H mice by ERG. Mice were injected with either (cell+gel), cells, or SHAM (pure saline water) at 4 weeks and tested weekly for cone retinal function with ERG apparatus. A. Photopic B-wave measured at 600 cd.s/m² for all groups. B. Percent increase calculation for Cell vs SHAM and Cell+Gel vs SHAM injections. (**p=0.0.00).

DETAILED DESCRIPTION

[0042] Described herein are methods and compositions that address limitations and drawbacks associated with previous approaches in the isolation, purification, and enrichment of rod progenitor cells. Provided are methods for producing enriched populations of human progenitor rod photoreceptor cells (hPRP) and EVs derived therefrom, as well as methods of using the hPRP and EVs to diagnose and treat retinal disease.

Human Progenitor Rod Photoreceptor Cells (hPRP)

[0043] Provided herein are purified or enriched populations of rod photoreceptor precursor cells, for example, human progenitor rod photoreceptor cells (hPRP), methods of producing these cells and use of the cells for the treatment of ocular disorders, e.g., retinal degenerative diseases, and other diseases. Described are methods for the isolation, purification and expansion of rod photoreceptor progenitor cells using a microfluidic based cell sorting approach, including a scalable GMP capable protocol for production, e.g., of human progenitor rod photoreceptors, providing the ability to create universal, allogenic, rod photoreceptor cells for preserving and restoring vision, e.g., as seen in FIG. 1.

[0044] As shown herein, the combination of CD73/CD276 is useful as a surface marker for labeling and isolating rod cells. CD276 was found to be upregulated in rod progenitor cells and can be released from the membrane and on the cell surface. We used it for labeling the rod progenitor cells (CD276 is exclusively expressed only by rod progenitor cells). CD73 is also known as 5'-nucleotidase ecto (NT5E); exemplary sequences for CD73 are provided in GenBank at Acc. Nos. NP_002517.1 (5'-nucleotidase isoform 1 preproprotein) and NP_001191742.1 (5'-nucleotidase isoform preproprotein). Exemplary sequences for CD276 are provided in GenBank at Acc. Nos. NP_001019907.1 (CD276 antigen isoform a precursor); NP_001316557.1 and NP_079516.1 (CD276 antigen isoform b precursor); and NP_001316558.1 (CD276 antigen isoform c).

[0045] Fetal retina cells (e.g., at 10-16 weeks in human embryonic age (fetal age) or an analogous age in development for other mammals such as mice), optionally cultured in hypoxic conditions (3-10% O₂), optionally with a nutrient mix containing serum (such as fetal bovine serum (FBS)) or a serum replacement (such as PHYSIOLOGIX Xeno-Free Serum Replacement (Nucleus Biologics) or KNOCKOUT Serum Replacement (Thermo Fisher)), EGF, and FGF as described herein, provide a timeframe during retinal development where the expression of rod progenitor cells is highest; differentiated iPSC and ESC can also be used. When iPSC/ESC are used, the ESCs/iPSCs are grown into optic cup or retinal organoids and (e.g., at days 40 to 60) the organoids

are used as a starting point for sorting using a method described herein, e.g., based on the presence of CD73 and CD276. See, e.g., Nakano et al., *Cell Stem Cell*. 2012 Jun. 14; 10(6):771-785; Kuwahara et al., *Methods Mol Biol*. 2017; 1597:17-29; Eiraku et al., *Nature*. 2011 Apr. 7; 472(7341):51-6; Reh and Fischer, *Methods Enzymol*. 2006; 419:52-73; Fathi et al., *Front Neurosci*. 2021 Apr. 20; 15:668857; and Achberger et al., *Adv Drug Deliv Rev*. 2019 Feb. 1; 140:33-50.

[0046] The present techniques can be used for isolation of rod progenitor cells, optionally using microfluidic devices such as the MACSQUANT TYTO sterile cartridge microfluidic cell sorting system (Miltenyi) to provide an enriched population of cells with higher purity and viability than previously reported. The microfluidic device allows capture of cells labeled with both positive markers (CD73/CD276) and/or deletion of the population of cells that was unwanted (all the rest of the cells that are CD73⁻ and/or CD276⁻). The present methods can be used to obtain populations of cells that are at least 75%, e.g., at least 80%, 0.5%, 90%, or 95%, cells that are positive for (express) both CD73 and CD276. In the present experiments, confirmatory markers Recoverin, CRX, CD73, and CD276 expression showed about 95% of positive for these markers. Furthermore, maintaining the isolated cells using culture conditions described herein allowed for cell proliferation, to obtain large number of hPRP. When transplanted into mouse eyes, the hPRP cells showed an extremely high capacity to provide neuroprotection in a degenerative model. To the best of the present inventors' knowledge, this is the first description of methods to isolate and culture a highly pure rod progenitor cells capable of offering high neuroprotection to the retina.

[0047] The hPRP cells can be differentiated to provide a population of human rod photoreceptor cells. A number of protocols are known; an exemplary protocol comprises culturing the cells in DMEM/F12 3:1+KSR/FBS with 0.1% 2-ME (b-mercaptoethanol), 0.2% IGF-1 (IGF1 Recombinant Human Protein), 2% B27 supplement, 1% taurine, and 1 μ M 9-cis retinal. The media is changed about every 2 days and the cells mature to rod photoreceptors in about 60 to 90 days.

Extracellular Vesicles (EVs) Derived from Human Progenitor Rod Photoreceptor Cells (hRP).

[0048] EVs and microvesicles are lipid enclosed cell fragments with diameters ranging from 50 nm to 2 μ m; they can be derived from almost any cell type, including embryonic stem cells, hematopoietic stems, neurons, and malignant cells. EVs typically have a diameter of 30-130 nm and are formed through the endosomal-sorting complex required for transport (ESCRT). Microvesicles are heterogeneous in size with diameters from 100-2000 nm. While EVs fall within the size range of microvesicles, microvesicles are distinct in formation, secretion, and content. The formation of microvesicles involves interactions between cytoskeletal and phospholipid proteins of the plasma membrane. EVs and microvesicles have been shown to be involved in cell-cell communication via transfer of proteins, mRNA, miRNA and DNA, e.g., as seen in FIG. 2. EVs and microvesicles encapsulate factors representative of cell of origin genotype and phenotype.

[0049] Provided herein are EVs derived from human progenitor rod photoreceptor cells (hPRPs) isolated using a method described herein. Some or all of the proteins shown in Table 1 are preferably present in EVs for use in the present methods and compositions, optionally including at least one, two, three four, five, or all seven of Thioredoxin (TXN, also known as RDCVF; NP_001231867.1); EGF containing fibulin extracellular matrix protein 1 (EFEMP1; NP_001034437.1); glutathione S-transferase Mu 1 isoform 1 (GSTM1; NP_000552.2 or NP_666533.1); sarcoglycan delta (SGCD; NP_000328.2, NP_758447.1, or NP_001121681.1); staphylococcal nuclease and tudor domain containing 1 (SND1; NP_055205.2), ADAM metallopeptidase domain 9 (ADAM9; NP_003807.1); and ezrin (EZR; NP_003370.2). Thus the EVs can comprise specific proteins comprising one, two, three, four, five, six, or all seven of TXN, EFEMP1, (GSTM1, SGCD, SND1, ADAM9, and EZR. Without wishing to be bound by theory, it is believed that these proteins provide a therapeutic effect. A summary and diagram of exemplary EVs is shown in FIG. 2. Antibody-directed targeting of EVs increases specificity of delivery of neuroprotective proteins, miRNA and other molecular cargo in the EVs to photoreceptors. This can be used to promote delivery of appropriate concentrations of neuroprotective cargo and

internalization by photoreceptors to activate restorative and regenerative processes. Methods for targeting of EVs for delivery of therapeutic agents has been successfully demonstrated in non-ocular tissues (see, e.g., Chen et al., *Front. Cell Dev. Biol.*, 8 Oct. 2021.

doi.org/10.3389/fcell.2021.751079.

[0050] hPRP cells secrete EVs into the culture media continuously, e.g., throughout the entire manufacturing and culturing process. The EVs can be obtained from the hPRP cells using known methods. An exemplary manufacturing method is summarized in FIG. 3. In this example, EVs secreted by hPRP cells are obtained from their culture media. Media is frozen and then ultracentrifuge to isolate EVs and discard debris, dead cells, and other undesirable components. The final pellet is resuspended in PBS, lyophilized and banked, keeping the EVs structure intact. Finally, the EVs are resuspended in an aqueous carrier (optionally comprising PBS or a hydrogel carrier) and injected in the vitreous of the eye. The media comprises the selected and secreted EVs. [0051] The EVs or hPRPs can be suspended, e.g., in a physiologically acceptable buffer such as phosphate buffer saline (PBS), or in a biocompatible polymer, e.g., a hydrogel. Since EVs injected simply in Phosphate buffer saline (PBS) may be targeted by the host immune system, thereby reducing their efficacy in inducing regeneration, polymeric scaffolds comprising biocompatible biomaterials can be used to encapsulate the EVs before delivery: the hPRPs can also be formulated in a polymeric gel for administration. Suitable bio-degradable and biocompatible polymeric materials can include one or more of gelatin, chondroitin sulphate, hyaluronic acid, alginate, collagen and chitosan-based scaffolds, e.g., gelatin-hyaluronic acid gel, e.g., a composition comprising gelatin hydroxyphenylpropionic acid (gelatin-HPA) and hyaluronic acid-tyramine (HA-Tyr) (see, e.g., WO2021113515 and Dromel et al., *NPJ Regen Med.* 2021 Dec. 20; 6(1):85). Such materials offer non-toxic and biocompatible degradation products while themselves triggering a higher immune response than synthetic polymer. Other materials that can be used includes but not limited to polycaprolactone (PCL), polylactic-glycolic acid (PLGA), polyethylene glycol (PEG), polylactic acid (PLA), polyetherimide (PEI), PNIPAAm, or hydroxy ethyl-methacrylate. See, e.g., US 20140231381 and US 20110004304.

Methods of Use

[0052] The EVs and hPRP produced as described herein can be used to treat retinal degenerative diseases, e.g., diseases associated with loss of photoreceptor rod or cone cells such as macular degeneration, e.g., dry macular degeneration; retinitis pigmentosa; and other Inherited Retinal Diseases, including Stargardt's Disease, Leber congenital amaurosis (LCA), or cone-rod dystrophy (CRD); in particular, the present methods can be used in conditions where rods are lost first. For example, certain retinal diseases affected by loss of cones (e.g., secondary loss of cones in retinitis pigmentosa (RP) leads to blindness) can be treated using hPRPs or EVs isolated from hPRP cultures as described herein; in some embodiments the EVs comprise rod-derived cone viability factor (RdCVF), an inactive thioredoxin secreted by rod photoreceptors that protects cones from degeneration (Aft-Ali et al., *Cell.* 2015 May 7; 161(4):817-32). Degeneration of retinal photoreceptors is also seen in the late stage of dry AMD, also known as geographic atrophy (GA). In some embodiments, the present methods can include administering a population of hPRPs to replace cells lost to degenerative disease. The inherited retinal diseases treatable by a method described herein include Bardet-Biedl syndrome, autosomal recessive; Choriorretinal atrophy or degeneration, autosomal dominant; Cone or cone-rod dystrophy, autosomal dominant; Cone or cone-rod dystrophy, autosomal recessive; Cone or cone-rod dystrophy, X-linked; Congenital stationary night blindness, autosomal dominant; Congenital stationary night blindness, autosomal recessive; Congenital stationary night blindness, X-linked; Leber congenital amaurosis, autosomal dominant; Leber congenital amaurosis, autosomal recessive; Macular degeneration, autosomal dominant; Macular degeneration, autosomal recessive; Ocular-retinal developmental disease, autosomal dominant; Optic atrophy, autosomal dominant; Optic atrophy, autosomal recessive; Optic atrophy, X-linked; Retinitis pigmentosa, autosomal dominant; Retinitis pigmentosa,

autosomal recessive; Retinitis pigmentosa, X-linked; X-linked; Usher syndrome, autosomal recessive; autosomal dominant retinopathy; autosomal recessive retinopathy; mitochondrial retinopathy, and X-linked retinopathy, as well as retinopathy associated with syndromic/systemic diseases. See, e.g., RetNet, the Retinal Information Network, available at sph.uth.edu/retnet/home.htm.

[0053] The methods generally include administering therapeutically effective amounts of hPRPs or EVs as described herein, e.g., by intraocular administration, e.g. by subconjunctival, intracameral, or intravitreal injection: see, e.g., Amo et al., *Prog Retin Eye Res.* 2017 March; 57:134-185; Yamada and Olsen, *Dev Ophthalmol.* 2016:55:71-83. For example, purified and lyophilized EVs can be stored at -80°C ., and can be resuspended, e.g., in the operating room, before administration, e.g., by vitreous injection, e.g., with a 31-gauge needle. The EVs can be resuspended in a physiologically acceptable buffer, e.g., PBS, or in a polymeric scaffold as described herein. Dissociated hPRP cells, e.g., in suspension, in a hydrogel (e.g., as described herein, e.g., a gelatin-hyaluronic acid gel, e.g., a composition comprising gelatin hydroxyphenylpropionic acid (gelatin-HPA) and hyaluronic acid-tyramine (HA-Tyr) (see, e.g., WO2021113515 and Dromel et al., *NPJ Regen Med.* 2021 Dec. 20; 6(1):85), or in a retrievable permeable capsule loaded with stem cells, can be delivered, e.g., through the retina after pars plana vitrectomy (PPV) or through a transscleral approach without vitrectomy (Hinkle et al., *Stem Cell Research & Therapy* volume 12, Article number: 538 (2021); Falkner-Radler et al., *The British journal of ophthalmology.* 2011; 95(3):370-5; Bhattacharya et al., *Curr Mol Biol Rep.* 2017 September; 3(3): 172-182.

[0054] Approximately 3 μg of lyophilized EVs were delivered in 3 μL per injection in rodent studies. Dosing concentration, quantity, and frequency to be used clinically can be determined based on animal and clinical trial data.

EXAMPLES

[0055] The invention is further described in the following examples, which do not limit the scope of the invention described in the claims.

Materials and Methods

[0056] The following materials and methods were used in the Examples below.

Cell Culture and Isolation Process of Rods

[0057] Human Fetus arrival: Fetal eyes arrive within 24 h in 2-degree box. Dissection was started directly in order to keep high viability.

[0058] Dissection and dissociation: Dissection of the retina was performed on all eyes. Retinas were then dissociated in papain for 30 min in the incubator. Cell suspension was then centrifuged and seeded in a T25 or T75 flask coated with fibronectin (depending on the number of cells).

[0059] Culture for 1-2 weeks: Cells were cultured in a 2D layer for 1-2 weeks using FDA approved media (no animal products; described below) in hypoxic conditions (5% O_2). Upon reaching confluence (about 10 million cells) cells were then passaged and centrifuged to start sorting.

[0060] Sort with CD73/CD276 (PE and APC): Cells were stained with CD73-PE and CD276-APC for 30 min in 2°C . After washing and preparation of the cell sorter, cells were sorted by gating about 20-30% of the positive population. Pre-sort and post-sort analyses were performed.

[0061] Culture to reach high yield: The final cell line was cultured for 4 weeks with the same media and flasks to obtain about 40 million cells for use in experiments or to create a cell bank.

Culture Media

[0062] DMEM/F12—500 ml [0063] Knockout serum (KSR)—55 ml-10% [0064] 100 $\times$ 1-Glutamax 5.5 ml-1% [0065] 100 $\times$ non-essential amino acid—5.5 ml-1% [0066] Sodium Pyruvate—5 ml-1% [0067] B-mercaptoethanol—5 μL -0.001% [0068] rhEGF (peprotech)—1000 μL (10 $\mu\text{g}/1\text{ ml}$)-0.2% [0069] rhFGF (peprotech)—500 μL (10 $\mu\text{g}/\text{ml}$)-0.1%

Eyecup Collection

[0070] 1. Eye globe was placed in buffer (HBSS/PBS) on a petri-dish and retina was carefully

teased out by removing the lens and vitreous fluid and without trace of RPE or ciliary body. [0071] 2. The retina was suspended in 10 ml papain solution for 30 mins at 37° C. to dissociate tissue into single cell suspension. [0072] 3. 20 ml of IBSS was added to the tube and the sample was centrifuged at 2000 rpm for 5 min. [0073] 4. The buffer was aspirated and the pellet re-suspended in 1 ml media, then plated on a T675 flask for further expansion

Preparing Cells for Sorting

[0074] 5. Cells were lifted from confluent flask using trypsin (1:6 in IBSS). [0075] 6. Cells were incubated with trypsin at 37° C. for 3 to 4 min. [0076] 7. Cells were collected in 50 ml tubes and around 40 ml media containing KSR was added to the cell suspension. [0077] 8. The suspension was centrifuged at 1200 rpm for 5 min at 15° C. [0078] 9. Supernatant was aspirated and the cell pellet resuspended in 200 µl of Miltenyi running buffer (cat no. 130-107-207).

Staining for Pure Rod Photoreceptors Sort

[0079] 10. Cells were incubated in CD73 PE (Cat no #130-095-183, Miltenyi), CD276 Antibody, anti-human, VioBlue® (Cat no #130-099-998) for 30 mins in ice. [0080] 11. After incubation the cells were re-suspended in 10 ml of HBSS and filter using 30 µm filter (Miltenyi). [0081] 12. The sample was then centrifuged at 300 g for 5 mins to wash off the antibody, [0082] 13. Meanwhile the TYTO cartridge was prepared by injecting filtered 1 ml TYTO buffer in the input chamber. The cartridge was pressurized to move buffer into the positive chamber and the negative chamber. [0083] 14. The buffer was removed from the input chamber to conclude cartridge prep. [0084] 15. Supernatant was then discarded and depending upon cell number cells were re-suspended in 5 to 10 ml (4 to 6 million cells in 5 ml and above 5 million in 10 ml) of TYTO buffer and injected into the input chamber of TYTO cartridge for sorting. [0085] 16. 100 µl of the cell suspension was placed in an Eppendorf for MACSQUANT analysis before running TYTO sorting.

Culturing Post-Sorting

[0086] 1. Using a fine tip pipet, cells in positive chamber were removed and chamber was flushed using HBSS. [0087] 2. 50 µl of the cell suspension was put in an Eppendorf for MACSQUANT analysis post-sorting. [0088] 3. The rest of the cell suspension was washed with 15 ml of HBSS by centrifuging at 300× for 5 mins. [0089] 4. Cell pellet was resuspended in culture media and plated on fibronectin coated T-25 or T-75 flask [0090] 5. Incubate at 37° C. in hypoxic conditions (5% O.sub.2). [0091] 6. Flask is maintained without media change for 48 hrs [0092] 7. 2 days post incubation change complete media. [0093] 8. Thereafter every 2 days fresh media was added until confluency was obtained [0094] 9. Confluent flasks (should be confluent in 7 to 10 days) were trypsinized and the cells were seeded on one t-75 for further expansion. [0095] 10. Media was changed every day until confluency was reached, in 3 to 4 days.

Cell Sorting and Analysis

Sorting Strategy

[0096] Cells analyzed with MACSQUANT before running the TYTO are set up as the control for running the sort. Markers of choice, CD73 and CD276 are analyzed with the MACSQUANT by tuning the voltages in order to put the desired population in the highest quadrant of the intensity. After this first gating strategy is performed, the TYTO cartridge is placed in the machine and the microfluidic flow is started. Cells are first pushed through the negative chambers in order to stabilize the pressure (around 150 hPa) but also the flow of cells in front of the lasers (40 ms between two cells). In order for the machine to know its velocity, a cell must be stained in at least two different fluorochrome, this way the machine can calculate the time it takes the cell to go from the first laser to the second one, hence measuring its velocity. As soon as both these variables are stable, lasers can be started, and fluorescent data starts to appear showing an approximate of 3000-5000 cells that are being pushed in front of the lasers. Each cell passes in front of the three lasers and its fluorescent intensity is measured and reported on the graph.

[0097] The same gating strategy that was used on the control on the MACSQUANT was reported on the live TYTO sorting machine and the sort was started. However, in order to properly capture

each cell, the valve speed and delay of action must be properly set up. Back scatter was fitted for each cell population (BSB, and BSV) flowing through the microfluidic device, and threshold was set up at 10^2 to 5.10^2 for both BSB and BSV. Threshold and gate channels for PE and VioBlue were centered on the highly fluorescent population, around 10^2 to 10^2 . Gating time was set up at ms with a pressure of 150 hPa. These settings can be altered depending on cell size, shape, granularity, intensity, stiffness, and concentration, which can be changed by analyzing the cell staining and population prior to starting the sort. Overall, the goal is to modify the parameters to obtain a high fluorescent population around 10^2 to 10^3 in both PE and VioBlue laser that can be sorted.

[0098] During the entire sorting process (usually about 2 hours) the percent of sorted, gated and positive cells was measured and reported as a function of time. At the end of the sort, the cartridge was removed from the machine and cells taken back to the cell culture for next steps.

Flow Cytometry

[0099] hPRPs (5×10^5 /mL in media) were trypsinized and cell pellet collected was processed for the phenotype then analyzed using a Flow Cytometry assay.

[0100] Flow Cytometry was performed using MACSQUANT flow cytometer (Miltenyi, San Diego) (100). Cells collected and fixed with paraformaldehyde at 4° C. for 15 min. Cells were then washed in wash buffer (BD Biosciences) and incubated, at room temperature, in block buffer (Pharmingen staining buffer with 2% goat serum) for 30 min. Blocked cells were seeded onto a flat bottom 96-well plate (treated, sterile, polystyrene, Thomas Scientific) and stained with conjugated primary antibodies (DAPI-Vioblue, CD73-PE, CD276-APC, Cone Arrestin-FITC, Blue opsin-FITC, Rhodopsin-FITC, CRX-APC, Recoverin-APC, Calbindin-FITC, PkCa-FITC, Bma3a-FITC, Ki67-APC, Thy1.2-APC, Vimentin-FITC, Cynoxin-APC, PAX6-APC, NRL-APC, Cmyc-FITC) overnight at room temperature. Primary antibodies were diluted in 200 μ L of antibody buffer (TBS, 0.3% Triton X-100 and 1% goat serum). After cells were washed three times for 15 min, secondary antibodies were goat-derived anti-rabbit and anti-mouse and diluted 1:200 in antibody buffer (Jackson ImmunoResearch Laboratory). Secondary antibodies were applied and left at room temperature for 3 h. light scatter and fluorescence signals from each well were measured using the MACSQUANT flow cytometer (2×10^5 events were recorded). The results were analyzed using the MACSQUAiNTify software (Miltenyi), For each primary antibody DAPI-positive single cell population was gated. The ratio of positive cells in the gated population was estimated in comparison with blank and species-specific isotype control.

Immunohistochemistry (IHC):

[0101] Live hPRPs grown in chamber slides and cryosection from Long Evans left eye were fixed with 4% paraformaldehyde in 0.1 M PBS (Irvine Scientific) at room temperature for 20 min. These fixed cells and sections were blocked and permeabilized with a blocking solution [(Tris-buffered saline (TBS), 0.3% Triton X-100 and 3% goat serum (Jackson ImmunoResearch Laboratories, West Grove, PA)] for 15 min. Samples were then rinsed twice with 0.1 M TBS buffer for 15 min each time, mounted on polysine microscope slides and incubated with primary antibodies overnight at 4° C. (DAPI-Vioblue, CD73-APC, CD276-Vioblue, Rhodopsin-Cy3, Recoverin-FITC, PKCa-Cy3, CRX-APC, Ki67-CV3, PAX6-FITC) at concentrations determined in laboratory. The next day, samples were rinsed three times with TBS for 15 min. Secondary antibodies (goat-derived anti-mouse and anti-rabbit) were applied for 1 h at room temperature. Samples were then washed one last time with TBS before being mounted on polysine microscope slide with 10 $\times$ viscosity slide mounting medium. Digital images were obtained with an Epifluorescent microscope using 20 $\times$ objective. Electronic image files were managed using Matlab software.

In Vivo Assessment of Cell and EV Effect

In-Vivo Transplantation

[0102] Fifteen rd-1 (C3H/HeOuJ) (age 3 weeks, approximate weight 15 g) were used as recipients

in the experiment. Transplantation was performed on non-immuno-suppressant mice. Mice were sedated intraperitoneal injection of ketamine (100-200 mg/kg) and xyalzine (20 mg/kg) for anesthesia. Eyes were first anesthetized using topical ophthalmic proparacaine (0.5%) followed by Genteal to keep the lens moist during the surgery.

[0103] Recipient mice were injected in sub-retinal space and intravitreal with hPRP or Rod-Exo single-cells injections. A conjunctival incision and a small sclerotomy were performed using a fine disposal scalpel. Cells were injected into the subretinal space using a glass pipette (internal diameter, 150 μ m) attached to a 50- μ L Hamilton syringe via a polyethylene tubing. The hPRPs were injected into the retina bleb as a single-cell suspension in PBS. All samples contained approximately 1×10^5 cells and the injection volume were 2 μ L for all replicates. Similarly, intravitreal injection was performed using 31 g needle with 3 μ L of cells in PBS. For sub-retinal injections using a glass coverslip applied on the eye checked bleb presence. Subretinal space injection was considered successful is a shining bleb was seen under the dissection surgical microscope. Triple antibiotic (Bac/Neo/Poly) was given locally at the end of the surgery to prevent further infection. The mice were then placed in their cages for a 21 day study.

[0104] The research protocol was reviewed and approved by the Schepens Eye Research Institute Animal Facility and was in accordance with the Association for Research in Vision Ophthalmology Statement for the Use of Animals in Ophthalmic and Vision Research.

Tissue Processing

[0105] 8-, 15- and 21-Days post transplantation rats were sacrificed by CO₂ suffocation for 2 min. Eyes were enucleated and placed in 4% paraformaldehyde for 24 hrs. Tissues were subsequently saturated with increase concentration of sucrose (5%, 10%, 20%) containing Sorensen phosphate buffer. Eyes were left in 30% sucrose overnight or until dissection. The tissues were embedded in cryo-section gelatin medium overnight and sectioned at 15 μ m thickness on a cryostat.

[0106] For subretinal injection, during the sectioning process, every 5 μ m section was stained and examined by epi-fluorescence for hPRPs presence with TRA-1-85-FITC and STEM121-FITC (human cells marker), rhodopsin-PE and recoverin-APC (host photoreceptor marker) and DAPI-Vioblue (cell nuclei). Sections were observed under Lecia Sp8 confocal microscope for engraftment and cells survival in both sub-retinal and intravitreal space of the mice.

[0107] For vitreous injection, during the sectioning process, every 6 μ m section was stained with H&E and examined under brightfield microscopy for morphology and architecture of the retina, along with photoreceptor layer thickness, Outer nuclear layer of photoreceptor was counted blindly to measure the potential neuroprotective effect of hPRPs. Frozen vials of media are kept at -80 C before use. Before performing isolation and lyophilization of EVs, frozen vials of media were analyzed by Dr. Redenti laboratory, where full proteomics and miRNA analysis was performed. Common protocol for EVs isolation and analysis was performed.

Rod-Exo Manufacturing Process

[0108] The manufacturing process is summarized in FIG. 3. EVs secreted by hPRP cells were obtained from their culture media. Media was frozen and then ultracentrifuged to isolate EVs and discard debris, dead cells, and other undesirable components. The final pellet was resuspended in PBS, lyophilized and banked, keeping the EVs structure intact. Finally, EVs were resuspended in water and injected in the vitreous of the eye. The hPRP secreted EVs throughout the entire manufacturing and culturing process. These EVs were released in the culture media. We saved and froze around 40-50 mL tube of this media after immunoselection (sort), and after passage 2. Those vials of media contain the selected and secreted EVs.

Rod-Exo Isolation from Media

[0109] Frozen media from fetal hPRP culture (for fetal-derived EVs) was thawed in a water bath (37 C). Following is a succession of ultracentrifugation of samples. First samples were spun at 300 g for 10 minutes in order to discard cells from the source. Following was a 2000 g spin for 10

minutes to discard dead cells and then a 10,000 g spin for 30 minutes to discard all cell debris. Finally, a double ultracentrifugation was performed to pellet down the EVs (only remaining component of the supernatant): both at 150,000 g for 70 minutes. Final supernatant was finally discarded, and a pellet of EVs was obtained. EVs were then measured, analyzed and used for testing and experiments.

Rod-Exo lyophilization and Preservation

[0110] Isolated pelleted EVs were resuspended in a final volume with PBS (50 mL).

Lyophilization, by controlled freeze-drying at -80°C ., was performed to obtain a final powdered product. This final powder was stored at -80°C . and was banked in vials containing 1 g of powdered EVs.

Testing of EVs and Proteins Functionality

[0111] Lyophilized EVs were resuspended in PBS and tested with Western blotting, Bradford assay, Nano sight, proteomics analysis and ELISA to check for concentration of proteins, of EVs, proteins functionality and presence of therapeutic proteins. Testing and analysis was compared to pre-lyophilized EVs using Western blotting for imaging and analysis of the presence of specific proteins; NANO SIGHT was used for analyzing and measuring the concentration of EVs; Bradford assays were used for measuring the total protein concentration; ELISAs were used for measurement of protein functionality; and proteomics analysis was used for a full analysis of fold change in proteins.

In-Vivo Testing

[0112] In the present experiments (Example 3), seven mice with the genotype $\rho^{-/-}$, aged three weeks and weighing approximately 120 g, along with P23 homozygous mice, aged four weeks and weighing approximately 117 g. were utilized. The mice were not given any immunosuppressant drugs before transplantation. To induce anesthesia, the mice were injected with ketamine (40=80 mg/kg) and xylazine (10 mg/kg). Proparacaine (0.5%) was topically applied to anesthetize the eyes, followed by Genteal to maintain the moisture of the lens during surgery. Recipient mice were injected in intravitreal space with HPRP (human progenitor rod photoreceptors), gel+HPRP and sham. Gel described is composed of gelatin-hyaluronic acid functionalized with tyramine side chains used at 5.5% (w/v) concentration (Dromel et al., NPJ Regen Med. 2021 Dec. 20; 6(1):85). The polymers were crosslinked using 0.1 U/ml of HRP and 1 mM of H.sub.2O.sub.2. Cells and gel were injected into the vitreal cavity using a glass pipette (internal diameter, 100 μm) attached to a 50- μL Hamilton syringe via a polyethylene tubing. All samples contained approximately 75,000 cells and the gel injection volume was 1 μL for all replicates. Triple antibiotic (Bac/Neo/Poly) was given locally at the end of the surgery to prevent further infection. The mice were then placed in their cages and returned to their normal habitat. The research protocol was reviewed and approved by the Schepens Eye Research Institute Animal Facility and was in accordance with the Association for Research in Vision Ophthalmology Statement for the Use of Animals in Ophthalmic and Vision Research.

Optometer Testing

[0113] The OptoMotry, designed for rodents by Cerebral Mechanics Inc., is used to perform Optokinetic Tracking (OKT) in a non-invasive manner. We placed mice on a platform surrounded by four LCD screens within a light-protected box, and presenting them with visual stimuli via the screens. To assess VA, mice were presented with vertical sine wave gratings of varying spatial frequencies. The gratings are displayed at a fixed distance from the animal, and the smallest spatial frequency that elicits a tracking response is recorded as the VA threshold. VA is measured separately for each eye, with an average testing time of 20 minutes per animal. For contrast sensitivity, tracking movements are identified as slow, steady head movements in the direction of the rotating grating. Spatial frequency thresholds are measured by testing the mice at various spatial frequencies between 0.064 and 0.514 cycles/degree, with an average testing time of 20 minutes per animal. The OptoMotry device uses a proprietary algorithm that adjusts the testing

stimuli based on the mouse's tracking reflexes. Contrast thresholds are measured at a spatial frequency of 02 cycles/degree and calculated as the reciprocal of the Michelson contrast. The reciprocal of the contrast threshold is then plotted for analysis.

Electroretinogram

[0114] To prepare for ERG testing, animals were anesthetized, and their pupils were dilated using a 1% Tropicamide eye drop, followed by a drop of 1% Proparacaine on the corneal surface. To prevent dehydration, a drop of Genteal (a corneal lubricant) was applied to the cornea of the untreated eye. In the treated eye, a drop of 0.9% sterile saline was applied on the cornea to allow electrical contact with the gold wire loop recording electrode. A 25-gauge platinum needle was inserted subcutaneously in the forehead to serve as the reference electrode, while another needle was inserted subcutaneously near the tail to serve as the ground electrode. ERG testing was done by series of flash intensities that is produced by a Ganzfeld, which is controlled by the Diagnosys Espion3. Both scotopic (low-light) and photopic (normal-light) responses were tested. During the test, a brief flash of light was presented to the eye, to stimulate the retina. The electrical response generated by the retina was recorded by the gold wire loop electrode on the cornea and transmitted to the Diagnosys Espion3. The response was analyzed to determine the animal's retinal function. This procedure can help diagnose and monitor various retinal diseases and disorders.

Example 1. Production of hPRP

[0115] Fetal retinas were cultured as described above in media and sorted based on expression of CD73+ and CD276+ with a Miltenyi TYTO cell sorter. Sorted cell numbers and morphology can be seen in FIGS. 4 and 5.

4.2 Sorting of hPRP from Fetal Tissue

[0116] FIG. 6 shows the live gating strategy and sorting of hPRP from fetal retina using both PE and VioBlue laser respectively for CD73 and CD276 positive cells. A gate is drawn around the correct population to start the sort and isolate pure hPRP. Post-sorting, all population are analyzed (input, positive and negative fractions). FIG. 7 shows the percent of positive hPRP in each population, reaching more than 95% in the positive chamber of the TYTO.

[0117] To confirm the success of the entire sort-run, analysis of trigger rate, sort rate, and percent of successfully sorted cells is monitored and analyzed throughout the entire process. FIG. 8. Shows the evolution of those variables during a 1 h sort. Finally, the final population of hPRP is analyzed in the MACSQUANT flow cytometer to precisely measure the number of positive cells (CD73+/CD276+) which were captured in the positive chamber compared to the input of fetal retina (FIG. 9).

4.3 Characterization of hPRP

[0118] After sorting and 4 weeks of culturing, hPRP were analyzed for their phenotype using both flow cytometry and immune-histochemistry staining (shown in FIGS. 10-13). FIG. 10 presents the critical marker for high purity of rods in hPRP (CD73, CD276 and recoverin). These markers were found to be higher than 95% in all cell population after culturing and sorting. Analysis of many retinal markers were performed to prove the presence of rods and the absences of all other retinal cells with flow cytometry (FIG. 11). The presence of high expression of rod markers confirmed the purity of hPRP, while all other retinal cell markers (RGC, bipolar, amacrine, horizontal, cones, and Muller cell markers) were found to be negative, assuring a high rod purity. To confirm these results, hPRP were stained the same marker and imaged under fluorescent microscopy to prove their expression (FIG. 12-13). High expression of rod and photoreceptor markers were found, while other retinal cell marker were absent from hPRP.

4.4 In Vivo Injection of hPRP In Vitreous and Subretinal Space

[0119] To prove the effect of hPRP in neuroprotection of the retina. Cells were injected in the vitreous and subretinal space of rd1 mice (a well-characterized retinal degeneration mouse model). Cells were injected at P25 and animals were sacrificed and analyzed 7 and 21 days post transplantation. To prove neuroprotection, retinas were stained with Cone Arrestin marker (in red)

to measure the amount of remaining host cones in the retina (which degenerate at a controlled rate). FIG. 14 shows the wholemount staining of these retinas showing the presence of a large number of cones at day 21, suggesting a high neuroprotection effect from the injection of hPRP. Retinas were also dissected in order to measure the outer nuclear layer (ONL) size, confirming the presence of photoreceptors in the degenerating rd1 mice. After measurement (FIG. 15) a larger number of cells in ONL were found for mice that received hPRP both in the subretinal space and vitreous compared to control. Finally, in order to test the safety of hPRP injection, intraocular pressure (TOP) was measured during the entire experiment, as shown in FIG. 16. A normal increase in IOP was found for a few days after vitreous injection, and went back to the normal range (10-20 mmHg) after 5 days with no increase for the rest of the experiment.

[0120] The sorting process (CD73.sup.+/CD276.sup.+) for purification of rod precursor photoreceptor cells yielded a viable 95% pure population of rod photoreceptor cells with a Ki-67 index of about 20% (showing a consistent proliferation).

[0121] These cells can be used for cell replacement, drug discovery and screening or other uses where photoreceptors or their precursors might be needed.

Example 2. Isolation and Characterization of Rod-EVs

[0122] Proteomics analysis of EVs was performed and fold change in proteins was compared to 14-weeks fetal retina. As seen in the Venn diagram in FIG. 17, thirty-one proteins were found to be exclusive to our EVs, while 361 were found in our EVs and in fetal retina.

[0123] FIG. 18 summarizes the entire proteome cargo found and analyzed in our EVs. Proteins were segregated into six major groups: [0124] 1. Retinal Specific: proteins involved in the development and regeneration of retina (cones, photoreceptors, ganglion cells, bipolar cells) [0125] 2. Cell protection: proteins involved in neuroprotection of cells [0126] 3. CNS development: proteins involved in the development of neural cells (neurons, axons, synapses) [0127] 4. Immune response: proteins involved in triggering an immune response and immune reaction. (cytokines, macrophages) [0128] 5. Common cell process: proteins found in all cells, involved in normal cell activity (adhesion, proliferation . . .) [0129] 6. EVs specific: proteins found on the surface of all EVs

[0130] Table 1 shows the major and most important proteins found in each category, with their protein effect, the gene associated to the protein and a description of their activity.

TABLE-US-00001 TABLE 1 Vesicle Proteomic Cargo Composition Neuro-retinal Retinal Protection TXN RDCVF secretion Protection EFEMP1/GSTM1 Protects from AMD (Total = 80) SRPX/CLIC4/SGCD Protects from RD/RP ADAM9/SND1 Cone survival factor RPN1 Protects from Stargardt's CALR Protects from CRVO CCT2 Protects from LCA IQGAP1 Protects from CNV DYNC1H1 Rod inner/outer segments Neuroprotection MFGE8/CD63 VEGF/Cell survival RALB/FAM129B Apoptosis suppression Tissue Regeneration GAPDH Regeneration CAV1 Retinal homeostasis Retinal Photoreceptor EZR Rhodopsin Development Development GNB4 Rod development (Total = 42) UGP2 Rods nuclear TPBG Rod bipolar attachment Retinal Development GNB1/GNB2 Opsin GNAI2 Muller opsins Retinal Engraftment ITGA5 Integrin binding VCAN HA binding BGN Chondroitin binding CNS Neural Development TENM4 Synapse genesis Development SH3BGRL3 Neurogenesis (Total = 51) Neural Process FGF2 Growth Factor RHOG Neurites growth Neural Maturation GALNTL5 Axon growth SEPT8 Synapse signaling Immune Cytokinesis LBP/EEF1A1 Cytokine response Response IGHM Antibodies (Total = 26) Programmed Cell Death TGFB1 Apoptosis Immune Signaling C6/CD99 Membrane attack complex A2MG CNS stress marker

In order to simplify the analysis of all proteins, subcategories were created in each of the six different groups established previously. Analysis of effect and activity of each protein was made with a protein database (Uniprot) and a gene database (GeneCard). FIG. 19 shows the fold change of all proteins classified into all the subcategories for the entire proteome cargo of our EVs. Of note is that 42 proteins were found to be for retinal specific development, 80 for cell protection, 51 for

CNS development and only 26 for immune response.

[0131] The first comparison of proteome cargo was performed between our EVs and the 14-weeks fetal retina EVs. This comparison enables us to characterize the specific beneficial therapeutics but more importantly the immune triggering protein. As seen in FIG. 20, most common cell processes and EVs have a similar expression in the fetal retina and our EVs. On average in retinal specific proteins when comparing fetal retina and our EVs. However, neuroprotective proteins are extremely upregulated in our EVs compared to the fetal retina. Of most importance is the presence of well-known proteins involved in a lot of failure in cell therapy (TGFB1) and immune response (JCHAIN). In the analysis we found that these proteins are expressed 10,000 times less in our EVs compared to the fetal retina. Of note is that other immune triggering proteins were found at same level in the fetal retina and in our EVs.

[0132] The final analysis performed in the proteome cargo of our EVs showed the proteins present only in our EVs but absent in the fetal retina. As seen in FIG. 21, 31 proteins were found to be absent in the fetal retina while being highly expressed in our EVs. Of note is that no immune triggering proteins were found to be exclusive to our EVs or absent from the fetal retina, which suggest that those immune triggering proteins are common proteins found in most stem cell therapies.

[0133] The total list of proteins found in the EVs (402) and analyzed as described above is provide in Table 2.

TABLE-US-00002 TABLE 2 List of all proteins found in hPRP-Exo with fold change Protein GN Rod-Exo Protein effect Protein Category F5GX11 PSMA1 1.10E+08 Lisosomes Axon development P09543 CNP 1.63E+08 Axon genesis Axon development A0A087WYF1 LAMA2 7.55E+07 Migration Axon development O14818 PSMA7 8.82E+07 Lisosomes Axon development C9JCK5 PSMA2 1.05E+08 Lisosomes Axon development G3V3U4 PSMA6 1.43E+08 Lisosomes Axon development A0A0A0MTC7 LAMA4 2.36E+08 Adhesion Axon development P11047 LAMC1 1.80E+08 Adhesion Axon development P07942 LAMB1 2.61E+08 Migration Axon development Q96CX2 KCTD12 9.61E+07 At synapses Axon development A2BDY9 HLA-A 4.48E+07 Antigen Cell adhesion G3V511 LTBP2 1.45E+07 Fibronectin Cell adhesion H0Y4I7 HLA-B 1.43E+08 Antigen Cell adhesion A0A087WZU5 TSPAN6 4.65E+07 cell-cell Cell adhesion interaction E7EV71 LTBP1 2.15E+08 Fibronectin Cell adhesion Q5SRN7 HLA-A 7.40E+07 Antigen Cell adhesion P35556 FBN2 3.92E+07 Binding Cell adhesion P18084 ITGB5 1.59E+08 Integrin Cell adhesion P35555 FBN1 3.92E+07 Binding Cell adhesion H3BQF7 IST1 1.37E+08 Cell process Cell adhesion Q92896 GLG1 1.38E+08 Binds fibroblast Cell adhesion E9PSH3 TSPAN4 1.15E+08 cell-cell Cell adhesion interaction K7ENU8 BCAM 5.83E+07 Adhesion Cell adhesion P60981 DSTN 5.83E+07 Actin binding Cell adhesion A0A0D9SF54 SPTAN1 1.21E+08 Cytoskeleton Cell adhesion Q07065 CKAP4 1.77E+08 Microtubules Cell adhesion Q8NG11 TSPAN14 4.69E+08 cell-cell Cell adhesion interaction P21589 NT5E 2.26E+08 Nociception Cell adhesion A0A2R8Y6L3 RPS10-NUDT3 1.47E+08 Hydrolase Cell adhesion P55001 MFAP2 1.50E+08 Microfibrils Cell adhesion Q9BUD6 SPON2 2.90E+08 Cytoskeleton Cell adhesion H0Y7V4 DNAH8 5.42E+08 Microtubules Cell adhesion P11021 HSPA5 6.68E+08 Proliferation, Cell adhesion apoptosis Q16853 AOC3 7.49E+08 Binding Cell adhesion A0A3B31ST1 TSPAN9 5.99E+08 cell-cell Cell adhesion interaction P07996 THBS1 8.55E+08 cell-matrix Cell adhesion interaction P46939 UTRN 1.31E+07 ECM adhesion Cell adhesion A0A087WXX2 ALDOB 7.69E+07 Actin removal Cell morphogenesis I3L161 PLSCR3 2.14E+08 Mitochondria Cell morphogenesis A0A0J9YX77 MGAM 1.45E+08 sucrose Cell morphogenesis degradation Q60FE5 FLNA 1.87E+08 Actin Cell morphogenesis B1AHL2 FBLN1 3.22E+08 Actin Cell morphogenesis Q14315 FLNC 2.69E+08 Actin Cell morphogenesis P23142 FBLN1 2.48E+08 Actin Cell morphogenesis P09525 ANXA4 1.89E+08 Secretion Cell morphogenesis P11233 RALA 9.33E+07 Cell division Cell morphogenesis P50995 ANXA11 6.05E+08 Cytokinesis Cell morphogenesis P55072 VCP 4.21E+08 Everywhere Cell morphogenesis Q71U36 TUBA1A

9.13E+08 Microtubule Cell morphogenesis Q9BUF5 TUBB6 1.27E+09 Microtubule Cell morphogenesis Q8WZ42 TTN 8.1E+08 Mitosis Cell morphogenesis Q13509 TUBB3 1.47E+09 Microtubule Cell morphogenesis P08133 ANXA6 2.85E+09 Secretion Cell morphogenesis P04350 TUBB4A 1.92E+09 Microtubule Cell morphogenesis P07437 TUBB 2.02E+09 Microtubule Cell morphogenesis P68371 TUBB4B 2.02E+09 Microtubule Cell morphogenesis A0A087WT27 PGM3 3.64E+07 Nucleotide Cell morphogenesis Q16658 FSCN1 7.41E+07 Actin Cell process P35580 MYH10 1.55E+08 Migration Cell process O75083 WDR1 8.24E+07 Actin binding Cell process E9PQH6 RHOC 1.91E+08 Support, motor Cell process H0YJM8 PSMB5 5.88E+07 Removes protein Cell process A0A0C4DGB6 ALB 2.89E+08 Binding Cell process C9JRL4 MDH1 4.38E+07 Cytosol Cell process metabolism P20618 PSMB1 1.03E+08 Removes protein Cell process P02794 FTH1 1.50E+08 Ion statis Cell process Q99805 TM9SF2 6.64E+07 Cell process Cell process F8W1R7 MYL6 1.80E+08 Migration Cell process P61225 RAP2B 1.01E+08 Regulator Cell process A0A494C0U1 TPP2 5.53E+07 Homeostasis Cell process Q6P163 APOC2 1.16E+08 Neutral Cell process F5H6E2 MYO1C 1.29E+08 Actin binding Cell process D6RHE2 EIF4E1B 3.98E+07 Ribonucleoprotein Cell process C9J4V0 RAB7A 1.32E+08 Regulator Cell process P29373 CRABP2 1.31E+08 Retinoid Cell process Q9BS26 ERP44 6.64E+07 Inside cells Cell process C9JZR2 CTNND1 9.19E+07 Actin Cell process J3KTM9 KPNB1 6.09E+07 Nuclear pore Cell process complex C9IY94 SEPT2 1.77E+08 Cell adhesion Cell process P20340 RAB6A 2.74E+08 Regulator Cell process P28070 PSMB4 1.82E+08 Removes protein Cell process P35579 MYH9 3.91E+08 Migration Cell process P10301 RRAS 8.45E+07 Actin Cell process Q7Z304 MAMDC2 2.07E+08 MAM Cell process B4DQU5 RAB11A 1.67E+08 Regulator Cell process P61026 RAB10 2.52E+08 Regulator Cell process P61006 RAB8A 2.74E+08 Regulator Cell process Q14764 MVP 5.71E+08 Vault Cell process Q8N6Y2 LRRC17 2.99E+08 Bone marrow Cell process control M0R0E8 ZNF791 3.65E+08 Zinc binding Cell process Q9H0U4 RAB1B 3.32E+08 Regulator Cell process F5H265 UBC 1.08E+09 Eucaryotic cell Cell process P02792 FTL 3.57E+08 Ion statis Cell process Q5JXB2 UBE2NL 4.14E+09 Protein folding Cell process P30041 PRDX6 2.43E+08 Reduce oxidative Cell stress stress regulators Q9BSK4 FEM1A 5.82E+07 Anti-inflammatory Cell stress regulators O15040 TECPR2 4.50E+07 Anti-inflammatory Cell stress regulators K7ELW0 PARK7 8.21E+07 Inhibitor Cell stress regulators Q9Y625 GPC6 5.83E+07 Growth factors Cell stress binding regulators P30101 PDIA3 2.06E+08 Stress marker Cell stress regulators F5GZS6 SLC3A2 9.69E+07 Membrane protein Cell stress regulators H7C3T4 PRDX4 1.70E+08 Reduce oxidative Cell stress stress regulators A0A024QZ42 PDCD6 2.01E+08 Stress marker Cell stress regulators P61224 RAP1B 4.73E+08 Regulator Cell stress regulators A0A0A0MSI0 PRDX1 2.54E+08 Reduce oxidative Cell stress stress regulators H7C5E8 TF 6.54E+08 Process Cell stress regulators Q8WUM4 PDCD6IP 6.84E+08 Stress marker Cell stress regulators P31689 DNAJA1 3.71E+07 Apoptosis Cell stress suppression regulators J3KTF1 HYOU1 5.83E+07 Protects oxidative Cell stress stress regulators P61204 ARF3 2.09E+08 Ribosomes Cell wall Biogenesis P04259 KRT6B 2.13E+08 Epidermal barrier Cell wall Biogenesis R4GMT0 ACTR1A 1.26E+08 Microtubules Cell wall Biogenesis P13645 KRT10 2.50E+08 Epidermal barrier Cell wall Biogenesis Q9NVM1 EVA1B 1.64E+08 Membrane protein Cell wall Biogenesis P04264 KRT10 1.20E+09 Epidermal barrier Cell wall Biogenesis O00560 SDCBP 1.42E+09 Genesis Cell wall Biogenesis K7ERI8 K7ERI8 1.24E+08 ATP Cellular metabolism P50395 GDI2 1.76E+08 ADP Cellular metabolism P01111 NRAS 1.02E+08 GTP Cellular metabolism P54687 BCAT1 6.03E+07 CNS Cellular metabolism Q96FQ6 S100A16 1.29E+08 Proliferation Cellular metabolism B5MD04 PRAME 4.10E+08 Cell proliferation Cellular metabolism M0QYF6 GPR108 2.34E+08 AAV transduction Cellular metabolism Q14697 GANAB 2.34E+08 Glycolysis Cellular metabolism P04075 ALDOA 3.91E+08 Fructose Cellular metabolism A0A0A0MS51 GSN 3.21E+08 Filaments Cellular metabolism Q9C0H2 TTYH3 2.00E+08 Calcium transfer Cellular metabolism P14618 PKM 1.17E+09 ATP Cellular metabolism P69891 HBG1 1.50E+10 Oxygen binding Cellular metabolism C9JNR5 INS 1.45E+10 Insulin

Cellular metabolism O43493-2 TGO LN2 1.73E+07 Trans-Golgi Cellular metabolism I3L2H4 HGS
1.18E+07 Signaling sp Cytokinesis A0A075B6N9 IGHM 8.22E+07 Antibody Cytokinesis
A0A075B6I0 IGLV8-61 1.34E+08 Antibody Cytokinesis O95865 DDAH2 1.77E+08
Vascularization Cytokinesis P68104 EEF1A1 4.09E+08 Cytokine Cytokinesis B9A064 IGLL5
3.20E+08 Antibody Cytokinesis P18428 LBP 2.31E+08 Cytokine response Cytokinesis
A0A0A0MS10 IGHV2-5 4.30E+08 Anybody Cytokinesis P01701 IGLV1-51 1.87E+08 Antibody
Cytokinesis A0A0B4J1T9 IGKV3-15 3.75E+08 Antibody Cytokinesis P01709 IGLV2-8 1.18E+08
Antibody Cytokinesis D6RD17 JCHAIN 2.46E+08 Antibody Cytokinesis Q13200 PSMD2
4.98E+08 Proteosome Exosome signaling Q5T8U2 RPL7A 1.08E+08 Ribonucleoprotein Exosome
signaling E9PJD9 RPL27A 1.37E+08 Ribonucleoprotein Exosome signaling B5MCW2 RPL3
1.01E+08 Ribonucleoprotein Exosome signaling QST7N0 RPL5 6.05E+07 Ribonucleoprotein
Exosome signaling Q02878 RPL6 5.84E+08 Ribonucleoprotein Exosome signaling H0YA55 ALB
8.36E+07 Binding Extracellular Matrix E7ENL6 COL6A3 1.62E+07 Collagen Extracellular
Matrix A0A087X0S5 COL6A1 8.36E+07 Collagen Extracellular Matrix P12111 COL6A3
1.16E+08 Collagen Extracellular Matrix D6RGG3 COL12A1 4.35E+08 Collagen Extracellular
Matrix Q15113 PCOLCE 8.82E+07 Collagen binding Extracellular Matrix P20908 COL5A1
1.28E+08 Collagen Extracellular Matrix B4DLR2 FAP 1.53E+08 Degradation Extracellular Matrix
B0V114 FLOT1 1.31E+08 Scaffold Extracellular Matrix P03956 MMP1 1.26E+08 Collagenase
Extracellular Matrix A0A0C4DFX3 EMILIN1 5.22E+08 Binding Extracellular Matrix Q76M96
CCDC80 1.06E+08 Cell adhesion Extracellular Matrix P02461 COL3A1 2.25E+08 Collagen
Extracellular Matrix P12110 COL6A2 5.65E+07 Collagen Extracellular Matrix Q96CG8 CTHRC1
2.69E+08 Collagen Extracellular Matrix P08253 MMP2 6.92E+08 Degradation Extracellular
Matrix P02452 COL1A1 1.04E+09 Collagen Extracellular Matrix A0A087WTA8 COL1A2
3.03E+09 Collagen Extracellular Matrix P02751-15 FN1 1.49E+10 Fibronectin Extracellular
Matrix P14209 CD99 1.08E+08 T cells Immune signaling G3XAJ6 RFTN1 9.86E+07 T and B
cells Immune signaling P13671 C6 1.85E+08 Attack cell Immune signaling membrane P01023
A2M 1.74E+08 CNS marker Immune signaling P11216 PYGB 8.88E+07 Antioxidant
Immunosuppression E9PNW4 CD59 4.69E+08 Lower MAC Immunosuppression P04083 ANXA1
1.17E+09 Immune system Immunosuppression P09382 LGALS1 1.20E+09 T cell death
Immunosuppression P35442 THBS2 4.56E+08 cell-cell Immunosuppression interaction O60814
HIST1H2BK 7.44E+08 Nucleus Immunosuppression Q08380 LGALS3BP 5.36E+09 T cell death
Immunosuppression P09960 LTA4H 6.64E+07 Protects Immunosuppression inflammation
Q9BVK6 TMED9 9.98E+07 Binding for rods Immunosuppression P50991 CCT4 8.90E+07 Actin
Molecular motor F8VR50 ARPC3 1.13E+08 Motility Molecular motor O15511 ARPC5 4.22E+07
Motility Molecular motor P07951 TPM2 1.28E+08 Actin binding Molecular motor Q01518 CAP1
1.42E+08 Actin Molecular motor E9PK52 EPB41L2 7.91E+07 Actin Binding Molecular motor
P52907 CAPZA1 9.1E+07 Actin Molecular motor P49006 MARCKSL1 2.53E+07 Actin
Molecular motor O15144 ARPC2 7.71E+07 Motility Molecular motor P67936 TPM4 2.22E+08
Actin binding Molecular motor E7ENZ3 CCT5 1.15E+09 Actin Molecular motor P35221
CTNNA1 1.40E+08 Actin Molecular motor X6RJP6 TAGLN2 2.95E+08 Actin Molecular motor
H0YCU9 TAGLN 1.04E+08 Actin Molecular motor Q9Y490 TLN1 1.07E+08 Actin Molecular
motor P12814 ACTN1 2.27E+08 Actin Molecular motor P07737 PFN1 6.40E+08 Actin Molecular
motor P40227 CCT6A 1.15E+08 Actin Molecular motor E9PP50 CFL1 6.61E+08 Actin Molecular
motor H3BT58 COTL1 1.18E+08 Actin Molecular motor P60709 ACTB 5.10E+09 Actin
Molecular motor P63261 ACTG1 5.10E+09 Actin Molecular motor P63267 ACTG2 4.45E+09
Actin Molecular motor F8WCJ1 EIF5A2 7.30E+07 mRNA binding mRNA Processing H0YH88
NAP1L1 1.59E+08 Nucleosome mRNA Processing A6PVH9 CPNE1 1.38E+08 DNA binding
mRNA Processing P16401 HIST1H1B 1.46E+08 Chromosomes mRNA Processing P16403
HIST1H1C 2.47E+08 Chromosomes mRNA Processing Q00610 CLTC 3.08E+08 Cell motor
mRNA Processing H7C4S4 FXR1 2.47E+08 translation mRNA Processing P62826 RAN

1.76E+08 transposon mRNA Processing K7BMV3 H3F3B 1.66E+08 Chromosomes mRNA Processing P62805 HIST1H4A 5.74E+08 DNA binding mRNA Processing Q16777 HIST2H2AC 1.26E+09 Chromosomes mRNA Processing Q96KK5 HIST1H2AH 1.08E+09 Chromosomes mRNA Processing C9J0D1 H2AFV 1.17E+09 Chromosomes mRNA Processing P23634 ATP2B4 5.10E+07 Transmembrane mRNA Processing P26641 EEF1G 5.83E+07 Neural retina Neural development Q6N022 TENM4 5.86E+07 Neurons and Neural development axons P60903 S100A10 2.26E+08 Cell Neural development differentiation P15144 ANPEP 4.64E+08 Neuropeptide Neural development Q9H2G4 TSPYL2 7.94E+07 Neural synapses Neural development FSH2D0 C1R 7.52E+07 Complement Neural development protein Q09666 AHNAK 5.82E+08 Differentiation Neural development Q9Y4F1 FARP1 5.16E+07 Dendrites Neural development P62158 CALM1 3.79E+08 Neuron migration Neural development A0A494C0G5 AGRN 6.63E+07 Post-synaptic diff Neural development O00468 AGRN 6.63E+07 Post-synaptic diff Neural development Q5H9A7 TIMP1 2.26E+08 Low proliferation, Neural development low mitosis P16035 TIMP2 4.32E+08 Low proliferation, Neural development low mitosis P60174 TPI1 2.55E+08 Motor neurons Neural development Q14195 DPYSL3 4.27E+08 Axon guidance Neural development R4GN98 S100A6 9.68E+08 Cell Neural development differentiation P31949 S100A11 8.69E+08 Cell Neural development differentiation P07355 ANXA2 7.35E+09 Growth of neurons Neural development Q7Z4T8 GALNTL5 Neural development C9JV02 SEPT8 1.01E+08 Axons Neural development Q9NZN4 EHD2 9.35E+07 Caveolae Neural maturation P23284 PPIB 3.58E+08 Neurons and Neural maturation axons C9K028 NME1 7.74E+07 Neurons and Neural maturation axons P80723 BASP1 1.83E+08 End of axons Neural maturation P18669 PGAM1 3.45E+08 Motor neuron Neural maturation H7C3J1 TSSK4 3.16E+08 Structural integrity Neural maturation P62937 PPIA 4.91E+08 Neurons and Neural maturation axons P84095 RHOG 1.73E+08 neurite growth Neural maturation P60953 CDC42 1.82E+08 Cell division Neuron process A0A087WUF6 FGF2 2.87E+07 FGF growth factor Neuron process P20073 ANXA7 1.72E+08 Synapse release Neuron process Q04917 YWHAH 1.86E+08 morphogenesis Neuron process P62258 YWHAH 4.05E+08 morphogenesis Neuron process P63104 YWHAZ 4.44E+08 Neurogenesis Neuron process P27348 YWHAQ 3.51E+08 morphogenesis Neuron process P63000 RAC1 4.74E+08 synapses, Neuron process dendrites P05023 ATP1A1 2.17E+07 Channel Neuron process O95084 PRSS23 1.10E+08 Metabolic Neuron process Q9P2B2 PTGFRN 3.16E+08 Axon's growth Neuron process Q92743 HTRA1 2.63E+08 Metabolic Neuron process Q5T123 SH3BGRL3 1.10E+08 Cell formation, Neuron process neurons Q96TA1 FAMI29B 1.02E+08 Apoptosis Neuroprotection suppression F8VNT9 CD63 2.09E+08 Cell survival Neuroprotection Q15084 PDIA6 9.78E+07 Membrane protein Neuroprotection P05121 SERPINE1 5.62E+07 Blood cloth Neuroprotection P22413 ENPP1 1.43E+08 Angiogenesis Neuroprotection O60568 PLOD3 4.97E+07 Disease modifier Neuroprotection A0A0U1RQF0 FASN 9.66E+07 Retinal protection Neuroprotection P14543 NID1 9.43E+07 Basal membrane Neuroprotection P06733 ENO1 6.72E+08 Hypoxia tolerance Neuroprotection P63313 TMSB10 1.86E+08 Organization of Neuroprotection skeleton Q14112 NID2 2.74E+09 Basal membrane Neuroprotection A0A087X1J7 GPX3 2.88E+08 Oxidative stress Neuroprotection Q02809 PLOD1 1.88E+09 Disease modifier Neuroprotection P10909 CLU 4.94E+09 Reduces immune Neuroprotection P08758 ANXA5 3.07E+09 Anticoagulant Neuroprotection Q08431 MFGE8 1.73E+09 VEGF Neuroprotection P01008 SERPINC1 5.55E+09 Anticoagulant Neuroprotection P17677 GAP43 2.00E+08 Neuron Neuroprotection remodeling P06744 GPI 5.33E+09 Neurotrophic Neuroprotection O75131 CPNE3 1.76E+08 RGC Photoreceptor development P07195 LDHB 3.34E+08 Glycolysis Photoreceptor development H0YCG2 LAMP2 5.87E+07 Inner segments Photoreceptor development P00338 LDHA 4.38E+08 Glycolysis Photoreceptor development E7EQR4 EZR 4.55E+08 Rhodopsin Photoreceptor development A0A2R8Y478 CD9 9.38E+08 Cell migration Photoreceptor development E9PIM6 THY1 1.44E+09 RGC Photoreceptor development P08670 VIM 1.57E+09 Muller cells Photoreceptor development E9PJK1 CD81 3.36E+09 Muller, RPE Photoreceptor

development P11279 LAMP1 5.83E+07 Inner segments Photoreceptor development P26022 PTX3
2.26E+08 Cytokines Pro-inflammatory P14174 MIF 2.41E+08 Cytokine response Pro-
inflammatory C9JR96 MME 1.74E+08 Diabetic Pro-inflammatory retinopathy P55083 MFAP4
1.75E+08 Diabetic Pro-inflammatory retinopathy P09486 SPARC 2.33E+08 Ocular diseases Pro-
inflammatory P61586 RHOA 6.93E+07 Axon retraction Programmed Cell death H7BZ94 P4HB
8.59E+07 Apoptosis Programmed Cell death H0Y3Z3 P4HB 1.10E+08 Apoptosis Programmed
Cell death P35908 KRT2 1.83E+08 Cornification Programmed Cell death Q15582 TGFBI
3.35E+08 Immune Programmed Cell death G3XAL9 SLC12A2 6.64E+07 RPE Retinal
development D6RBT0 SEC31A 3.05E+06 Nuclear pore Retinal development complex R4GN83
BSG 2.31E+08 Retinal neuron Retinal development O14672 ADAM10 2.81E+08 Early retina
Retinal development E5RFU4 DPYSL2 5.05E+07 RGC Retinal development D6RFI1 DBN1
5.20E+07 Whole retina Retinal development P04899 GNAI2 4.32E+08 Muller glia Retinal
development Q9UBI6 GNG12 2.71E+08 Axonal guidance Retinal development P98160 HSPG2
6.62E+07 Basement Retinal development membrane P36955 SERPINF1 1.25E+08 RPE Retinal
development C9JIS1 GNB2 5.61E+08 Axonal guidance Retinal development Q9HAV0 GNB4
5.75E+08 Rods Retinal development P62873 GNB1 8.07E+08 Axonal guidance Retinal
development P26038 MSN 4.86E+08 Structure Retinal development P61769 B2M 4.06E+08
Neural retina Retinal development C9J419 PXDN 2.50E+08 Embryo Retinal development Q92626
PXDN 4.28E+08 Embryo Retinal development K7ES78 CAPNS1 5.83E+07 Embryonic dev
Retinal development C9JWG0 UGP2 1.13E+08 Rods nuclear Retinal development Q13641 TPBG
1.40E+08 Rod bipolar cells Retinal development P48061 CXCL12 3.65E+08 Cell proliferation
Retinal development P13612 ITGA4 6.83E+07 Integrin Retinal Engraftment fibronectin
A0A494C194 PEPD 1.21E+08 Collagen binding Retinal Engraftment P08648 ITGA5 1.63E+08
Integrin Retinal Engraftment fibronectin P06756 ITGAV 3.39E+08 Integrin laminin Retinal
Engraftment P50454 SERPINH1 2.92E+08 Collagen binding Retinal Engraftment H0YDX6 CD44
3.80E+08 HA binding Retinal Engraftment Q86W61 VCAN 2.16E+08 HA binding Retinal
Engraftment P05556 ITGB1 3.39E+08 Integrin collagen Retinal Engraftment E9PF17 VCAN
1.91E+08 HA binding Retinal Engraftment P21810 BGN 1.92E+08 Chondroitin Retinal
Engraftment sulfate O43854 EDIL3 3.81E+08 Binding Retinal Engraftment P10809 HSPD1
2.54E+07 Protection Retinal protection Q580Q6 EFEMP1 1.73E+09 Protects from Retinal
protection AMD H7C597 SND1 2.69E+09 Photoreceptor Retinal protection protection P10599
TXN 4.07E+09 RDCVF Retinal protection F8WF32 RPN1 6.16E+07 Protects from Retinal
protection Stargardt's O60701 UGDH 9.89E+07 Protects RGC Retinal protection A0A286YFA2
PHGDH 3.64E+07 Protects from Retinal protection MacTel P78539 SRPX 1.21E+09 Protects from
RP Retinal protection P07093 SERPINE2 5.83E+07 Angiogenesis Retinal protection P63092
GNAS 4.12E+08 RGC and ON Retinal protection P07900 HSP90AA1 3.71E+08 Protection
Retinal protection K7EJB9 CALR 1.15E+09 Protects from Retinal protection CRVO P08238
HSP90AB1 4.08E+08 Protection Retinal protection P07585 DCN 2.52E+08 Retinal structure
Retinal protection J3QKS4 TSG101 1.44E+09 Protects RPE Retinal protection P11142 HSPA8
6.27E+08 Protection Retinal protection Q14204 DYNC1H1 5.83E+07 Rod inner/outer Retinal
protection P09211 GSTP1 4.52E+08 Protects RPE Retinal protection A0A0J9YXZ5 IQGAP1
1.24E+09 Protects from Retinal protection CNV H3BLV0 CD55 1.36E+09 Protects from Retinal
protection AMD H3BRM6 GSTM1 1.77E+09 Protects from Retinal protection AMD Q9Y696
CLIC4 1.92E+09 Protects from RD Retinal protection Q13443 ADAM9 4.98E+09 Protects from
Retinal protection CRD A0A2R8Y484 CD47 8.20E+09 Cone survival Retinal protection F8VQ14
CCT2 8.44E+09 Protects from Retinal protection LCA Q92629 SGCD 5.95E+10 Protects from
Retinal protection AMD E9PL09 RPS3 7.64E+07 40S subunit Ribosome Biogenesis Q5T6W2
HNRNPK 6.38E+07 Ribonucleoprotein Ribosome Biogenesis P25398 RPS12 1.36E+08 40S
subunit Ribosome Biogenesis J3KSS0 RPL26 6.23E+06 Ribonucleoprotein Ribosome Biogenesis
G3V1B3 RPL21 5.83E+07 Ribonucleoprotein Ribosome Biogenesis B7Z645 SYNCRIP 6.21E+07

Ribonucleoprotein Biogenesis M0QZC5 RPS11 1.29E+08 40S subunit Ribosome Biogenesis H7C2W9 RPL31 1.34E+08 Ribonucleoprotein Ribosome Biogenesis H0YEN5 RPS2 1.43E+08 40S subunit Ribosome Biogenesis M0QZN2 RPS5 2.34E+08 40S subunit Ribosome Biogenesis F8W7C6 RPL10 7.64E+07 Ribonucleoprotein Ribosome Biogenesis J3JS69 RPS18 1.29E+08 40S subunit Ribosome Biogenesis C9J9K3 RPSA 1.64E+08 40S subunit Ribosome Biogenesis P05387 RPLP2 5.37E+07 60s subunit Ribosome Biogenesis D6RBD0 GNB2L1 1.61E+08 40S subunit Ribosome Biogenesis J3KMX5 RPS13 1.82E+08 40S subunit Ribosome Biogenesis P62701 RPS4X 1.02E+08 40S subunit Ribosome Biogenesis P62851 RPS25 1.15E+07 40S subunit Ribosome Biogenesis M0QX76 RPS16 2.40E+08 40S subunit Ribosome Biogenesis D6RGE0 RPS3A 3.2E+08 40S subunit Ribosome Biogenesis P05386 RPLP1 6.90E+08 60s subunit Ribosome Biogenesis ESRIP1 RPS20 7.53E+07 40S subunit Ribosome Biogenesis Q5JR95 RPS8 2.36E+08 40S subunit Ribosome Biogenesis P13639 EEF2 4.21E+08 mRNA Ribosome Biogenesis H7BYG8 LTN1 7.39E+09 Ribonucleoprotein Ribosome Biogenesis P09936 UCHL1 1.21E+08 Stroke treatment Tissue Regeneration Q9NZM1 MYOF 1.03E+08 Repair membrane Tissue Regeneration P14625 HSP90B1 2.47E+08 neural chaperon Tissue Regeneration P29966 MARCKS 8.18E+08 Neurons and Tissue Regeneration axons J3QSU6 TNC 7.23E+08 Neuron guidance Tissue Regeneration P04406 GAPDH 2.29E+09 Regenerate Tissue Regeneration Q15063 POSTN 1.10E+09 Stem neuron Tissue Regeneration proliferation B1ALD9 POSTN 1.10E+09 Stem neuron Tissue Regeneration proliferation B1AHC9 XRCC6 DNA repair Tissue Regeneration P00742 F10 7.20E+07 Cell migration Tissue Regeneration C9JKI3 CAV1 1.59E+08 In the retina, Tissue Regeneration homeostasis

Example 3. In Vivo Evaluation of/Rod-EVs

[0134] Rho^{-/-} mice are knockout mice that lack the gene for rhodopsin, a light-sensitive protein found in the rods of the retina. Rhodopsin is essential for vision in low light conditions. Therefore, Rho^{-/-} mice have impaired vision in dim light, and their rods do not function properly. They are commonly used in research related to retinal degeneration. P23h homozygous mice, on the other hand, have a mutation in the rhodopsin gene that causes it to produce a defective form of rhodopsin protein. This mutation is similar to the one found in humans with autosomal dominant retinitis pigmentosa (RP). P23h homozygous mice have progressive retinal degeneration and vision loss, similar to RP patients. They are used in research related to understanding the molecular mechanisms underlying RP and for testing potential treatments. So, while both rho^{-/-} and P23h homozygous mice are used in vision-related research, their genetic modifications and associated characteristics are different.

[0135] Due to this mutation, rod photoreceptors malfunction and start to degenerate as soon as 3 weeks post-natal. This is followed by the death of cones starting at 5-6 weeks post-natal. To neuroprotect the retina, we injected 4 week old mice with the technology cell+gel (gel composed of gelatin and hyaluronic acid combined at 5.5% w/v concentration and crosslinked using HRP and H.sub.2O.sub.2) to potentially halt and slow down the degeneration of these mice, with a focus on cone neuroprotection. We tested the functional and retinal behavior of these mice each week post-injection. The functional behavior testing, FIGS. 22A-B, was performed using optomotor testing to measure both visual acuity and contrast sensitivity. We confirmed, as shown in previous papers (Xiao et al., Invest Ophthalmol Vis Sci. 2019 Oct. 1; 60(13):4196-4204), that visual acuity in Rho^{-/-} does not decrease prior to 12 weeks, independent of the injection group. However, we observed a decrease in contrast sensitivity for SHAM group as soon as 2 weeks post-injection. The cell+gel treatment was able to stop that decrease and delay it by at least 1 month as seen in the significant differences at weeks 8 and 9. While this is an ongoing study, we expect to obtain high durability in neuroprotecting the functional behavior of these mice until at least 12-14 weeks.

[0136] These results were confirmed by the cone function test performed using electroretinogram (ERG), as seen in FIGS. 23A-C. We focused on Photopic B-wave, which is representative on cone function in the retina. As seen in FIG. 23A, a significant increase in cone function was observed at

7- and 8-weeks following injection, which was also measured using the percent increase (FIG. 23B) and confirmed with the ERG traces (FIG. 23C).

[0137] Overall, this ongoing study suggests that our gel+cell can neuroprotect the function of cones in the retina of degenerative Rho^{-/-} mice.

[0138] As stated in Methods, P23H have a genetic mutation with a misfolding of rhodopsin protein. We performed a similar study as for Rho^{-/-}, knowing that P23H have a faster degenerative profile. While no Scotopic wave was observed in P23H, we were able to measure, using ERG, the photopic B wave of 3 different groups (SHAM, cells, cell+gel) as seen in FIGS. 24A-B. A significant improvement in Photopic B-wave was observed as soon as 1 week after injection for both cell and cell+gel compared to SHAM group.

[0139] These results show the efficacy of this treatment even at a late-stage degenerative retina.

Other Embodiments

[0140] It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

Claims

1. A method of providing a population of progenitor rod photoreceptor cells (PRPs), the method comprising: providing an initial population comprising retinal cells from a mammal, preferably a fetal mammal; isolating cells expressing CD73 and CD276 from the initial population to provide a substantially purified population of PRPs; and optionally culturing the substantially purified population.
2. The method of claim 1, wherein isolating cells expressing CD73 and CD276 comprises using fluorescence activated cell sorting (FACS) or Magnetic-activated cell sorting (MACS), optionally in a microfluidic device.
3. The method of claim 1, further comprising culturing the initial population comprising retinal cells before isolating cells expressing CD73 and CD276.
4. The method of claim 1, wherein the substantially purified population of PRPs is cultured in media comprising serum or serum replacement, EGF, and FGF.
5. The method of claim 1, wherein the substantially purified population of PRPs is cultured in hypoxic conditions.
6. The method of claim 1, wherein the initial population comprising retinal cells is from a fetal mammal.
7. The method of claim 1, wherein the mammal is a human.
8. A substantially purified population of PRPs produced by the method of claim 1, preferably wherein at least 75% of the population comprises cells expressing CD73 and CD276.
9. (canceled)
10. A method of treating a subject who has a condition associated with loss of retinal rod photoreceptors, the method comprising administering to the subject a therapeutically effective amount of PRPs obtained by the method of claim 1, preferably wherein at least 75% of the population comprises cells expressing CD73 and CD276.
11. A method of providing extracellular vesicles (EVs) from rod photoreceptor progenitor cells, the method comprising: providing the substantially purified population of PRPs of claim 8; maintaining the substantially purified population of PRPs in culture in media, preferably wherein the media is in contact with the PRPs in the culture for at least 12, 18, 24, 36, 48, or 72 hours; removing the media from the culture; and isolating EVs from the media.
12. The method of claim 11, wherein isolating EVs comprises ultracentrifuging the media to obtain a pellet comprising EVs.

- 13.** The method of claim 11, further comprising lyophilizing the isolated EVs.
 - 14.** A composition comprising EVs obtained by the method of claim 11.
 - 15.** The composition of claim 14, wherein the EVs are in a polymeric scaffold.
 - 16.** The composition of claim 15, wherein the polymeric scaffold comprises polycaprolactone (PCL), polylactic-glycolic acid (PLGA), polyethylene glycol (PEG), polylactic acid (PLA), polyetherimide (PEI), PNIPAAm, or hydroxy ethyl-methacrylate.
 - 17.** The composition of claim 14, for use in treating a subject who has a condition associated with loss of retinal rod or cone photoreceptors.
 - 18.** A method of treating a subject who has a condition associated with loss of retinal rod or cone photoreceptors, the method comprising administering to the subject a therapeutically effective amount of PRPs obtained by the method of claim 11.
 - 19.** The method of claim 10, wherein the condition associated with loss of retinal rod or cone photoreceptors is an inherited retinal degenerative disease (IRD), optionally cone-rod dystrophy, retinitis pigmentosa, or Stargardt's disease; or macular degeneration.
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